Deploy Network in GNS3

1. Primeros pasos:

Primero necesitamos tener instalado en nuestro ordenador el programa GNS3 desde la página web oficial: https://gns3.com/software/download

Una vez descargado e instalado GNS3 en nuestro equipo, debererermos escoger el modo de trabajar con él. Debemos definir de que forma levantar el "Main Server" de GNS3.

Tenemos 3 opciones:

- Escritorio
- Máquina virtual
- Web

En este caso trabajaré con Máquina Virtual, ya que de este modo tendremos controlados los recursos (RAM, CPU y Disco) que se usan de forma más eficiente.

Para trabajar con la máquina virtual necesitaremos un gestor de máquinas virtuales o Hypervisor. Existen varios: Hyper-V, VirtualBox, VMware Workstation, Proxmox, entre otros.

En este caso utilizaremos el Hypervisor propiedad de Broadcom: ***VMware Workstation 17.6.4 build-24832109***.

Enlace de descarga (Deberemos registrarnos e iniciar sesión primero): https://support.broadcom.com/group/ecx/productfiles?subFamily=VMware%20Workstation%20Pro&displayGroup=VMware%20Workstation%20Pro%2017.0%20for%20Windows&release=17.6.4&os=&servicePk=533272&language=EN&freeDownloads=true

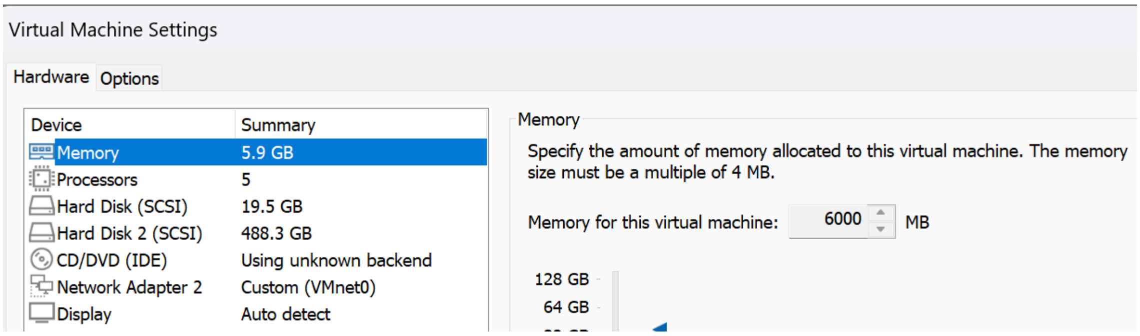
Una vez descargado el Hypervisor lo instalamos y ya podemos empezar a trabajar.

2. Levantar Main Server en Hypervisor y conectarlo con GNS3.

Ahora que tenemos instalado todo lo necesario, importamos la VM del Main Server en VMware Workstation y configuramos las características de la máquina.

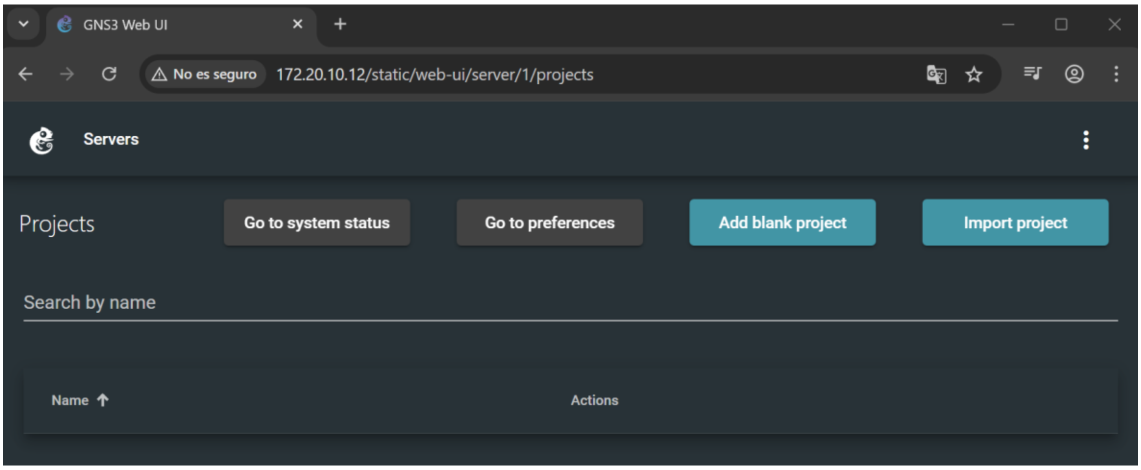
Podemos asignar los recursos que queramos, en este caso asignamos:

- 6 GB de RAM.
- 5 procesadores.
- 1 adaptador de Red en modo Bridge.



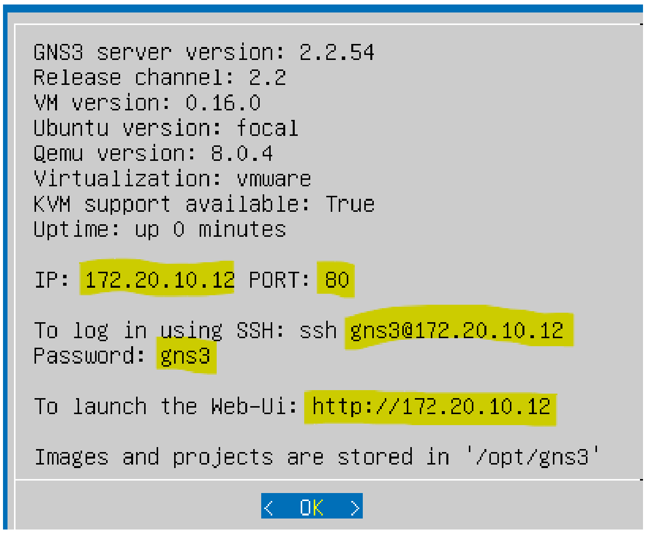
Una vez asignados los recursos encendemos la máquina virtual.

Probamos la conexión haciendo ping del Host (PC Anfitrión) al Main Server y intentamos navegar a la web del Main Server.



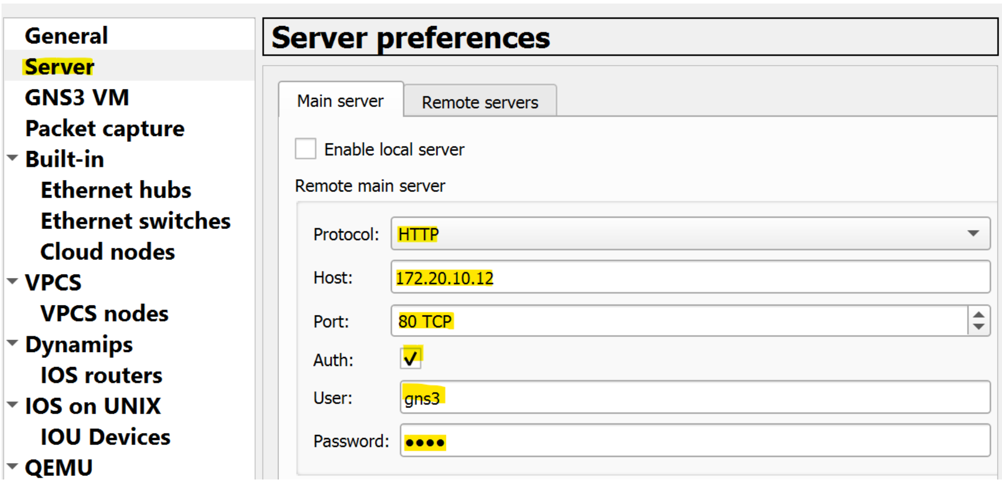
Ahora que comprobamos que podemos ver la web, procedemos a conectar el Main Server con GNS3.

Prestamos especial atención a la siguiente información que aparece en la VM:

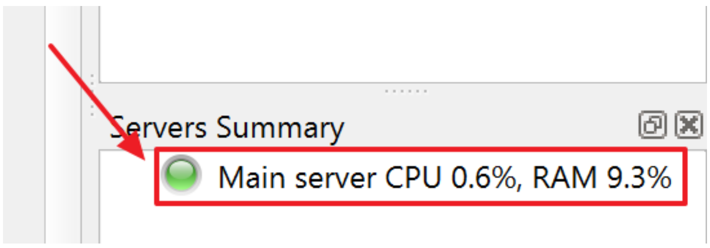


Ahora procedemos a abrir GNS3 >> Edit >> Preferences >> Server y deshabilitamos "Enable local server".

Después especificamos: Protocol, Host, Port, Auth, User y Password.

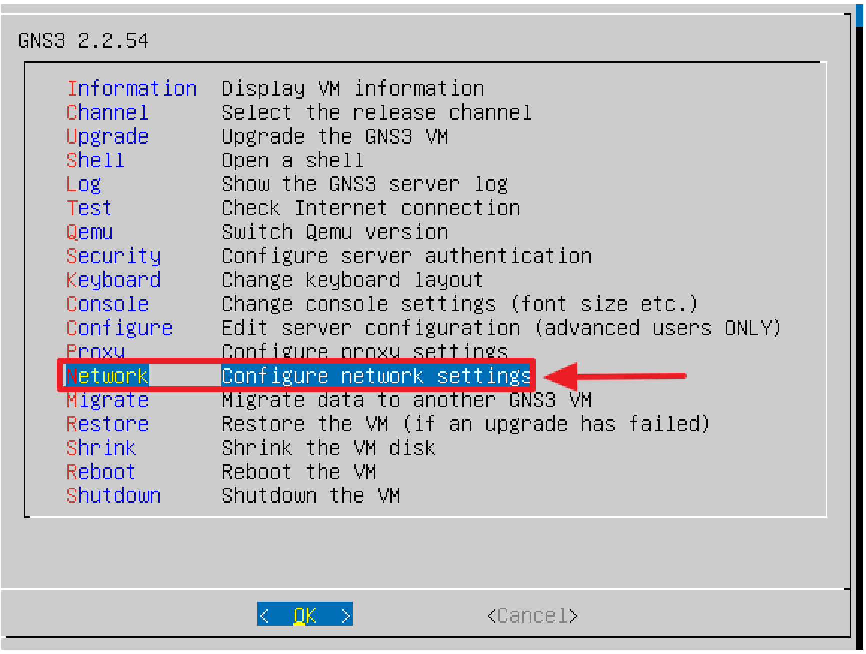


Si hemos configurado las cosas bien, nos deberá aparecer en la Columna "**Servers Summary**" el Main Server en verde.



Después de comprobar que el Main Server se conecta a GNS3 sin problemas, procedemos a fijar la IP.

Accedemos al menú principal en el Main Server y escogemos la opción “Network”. El servidor se reiniciará, aceptamos.



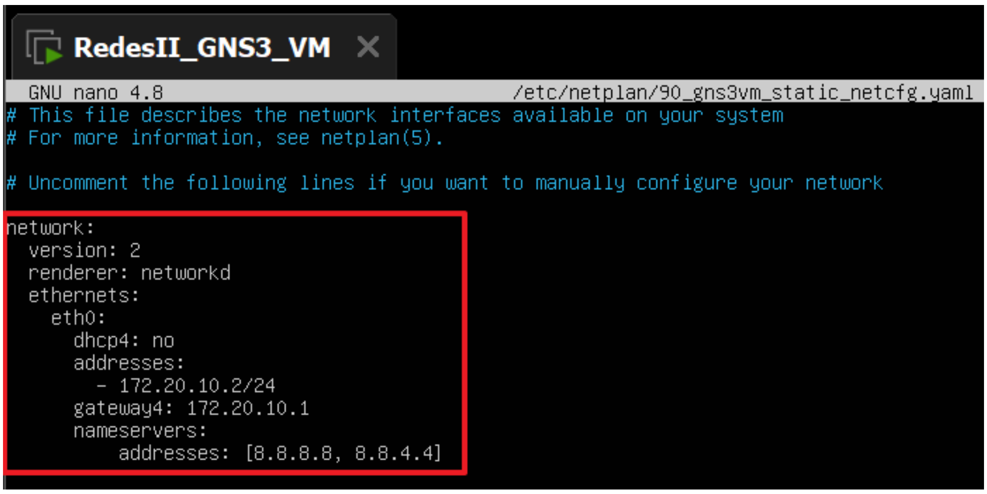
Configuramos la interfaz de red con la IP estática y DNS siguientes:

DHCP4: no

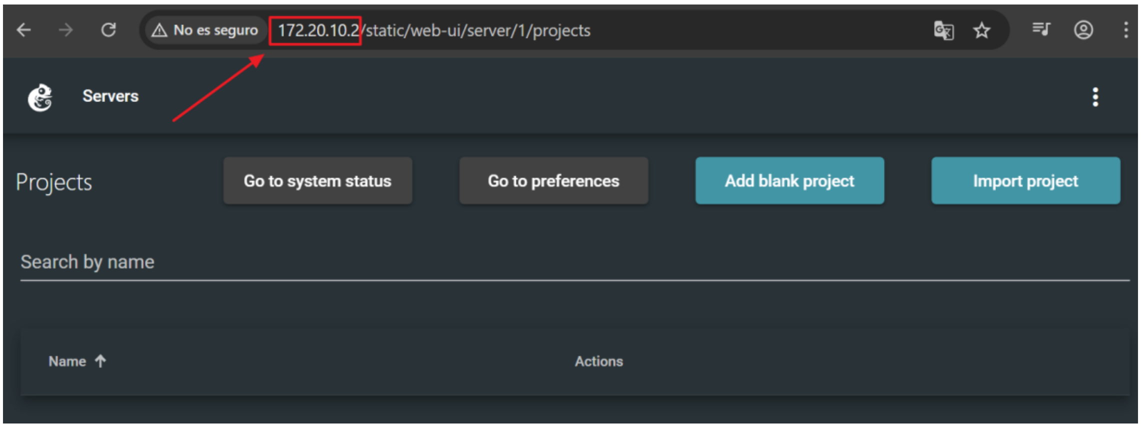
Dirección IP: 172.20.10.2/24

Gateway4: 172.20.10.1

Nameservers: 8.8.8.8, 8.8.4.4

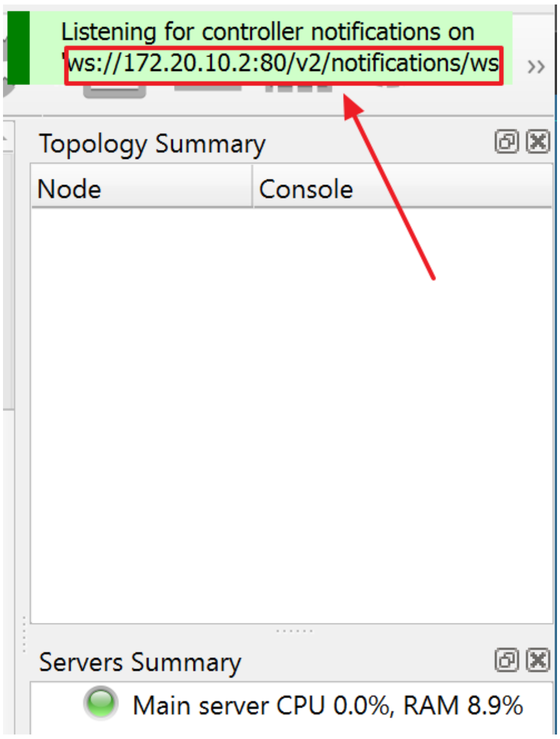


Comprobamos mediante ping y web que podemos ver el Main Server con la IP fija:

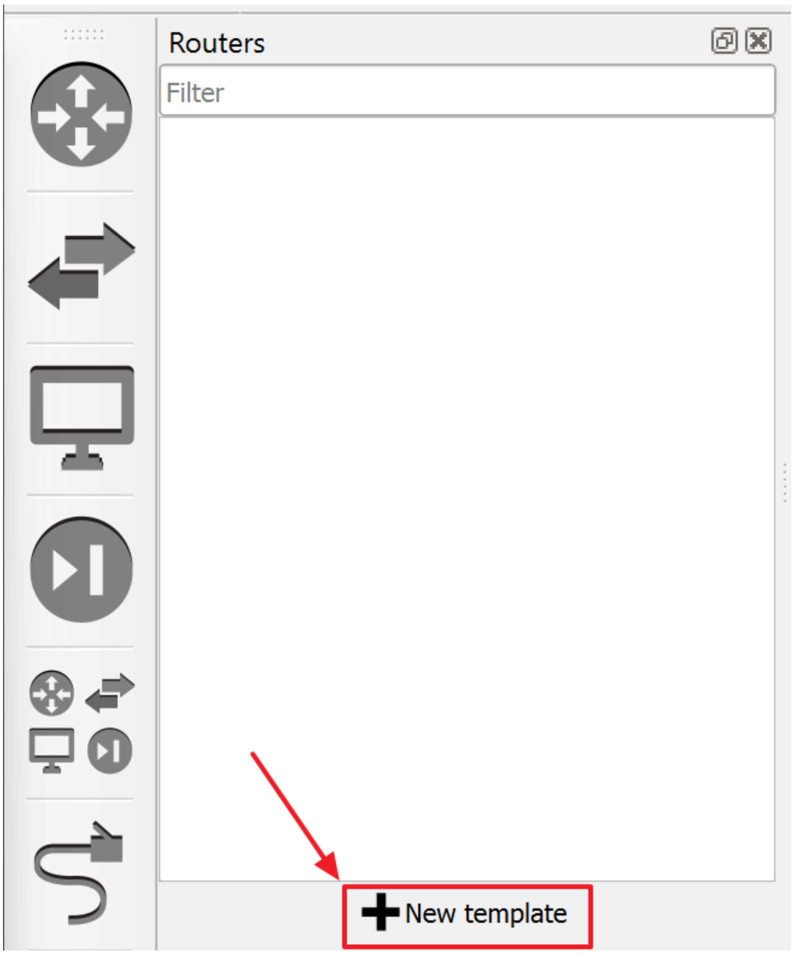


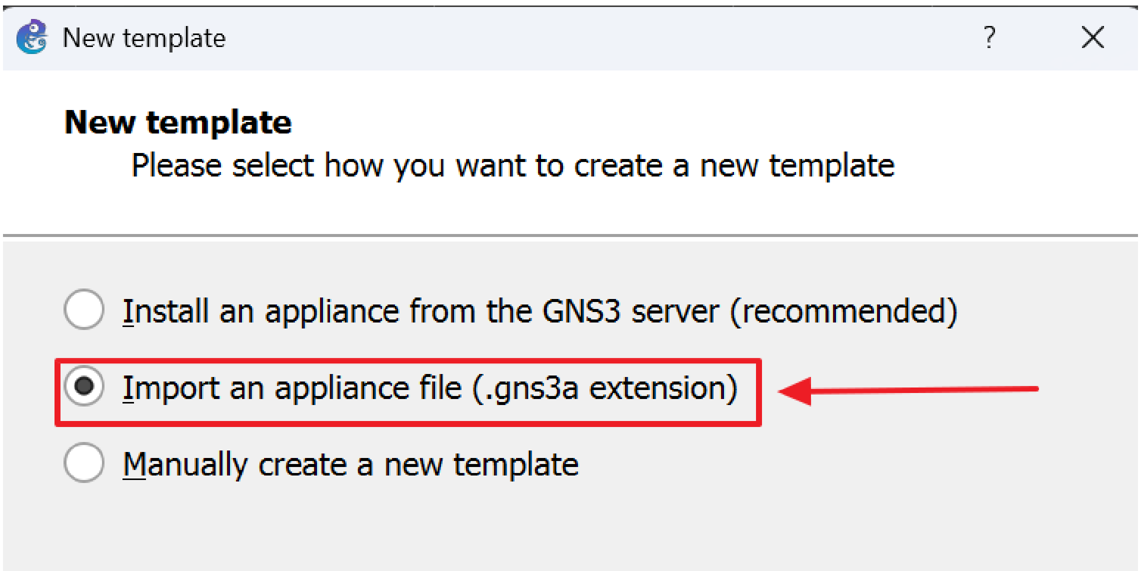
Ahora volvemos a GNS3 y cambiamos la IP del Main Server.

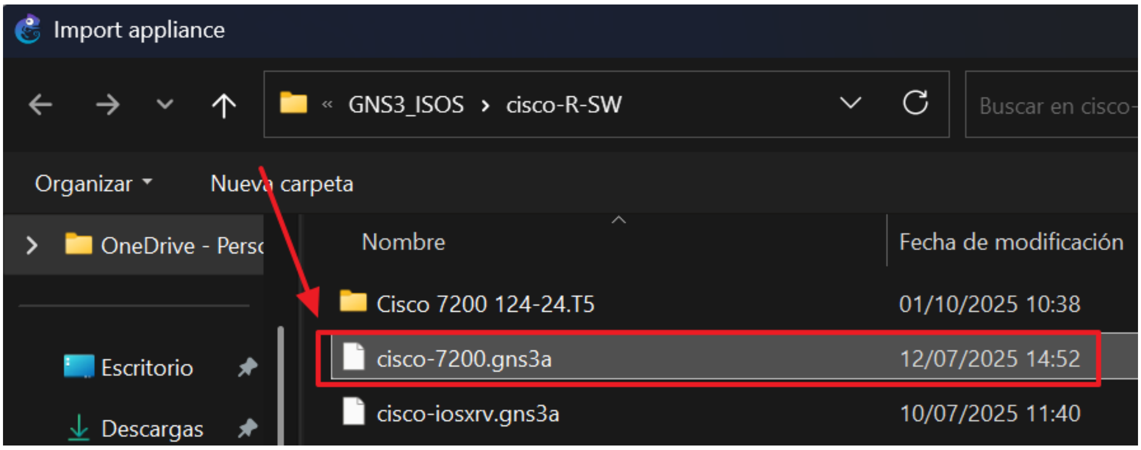
Comprobamos que funciona correctamente:

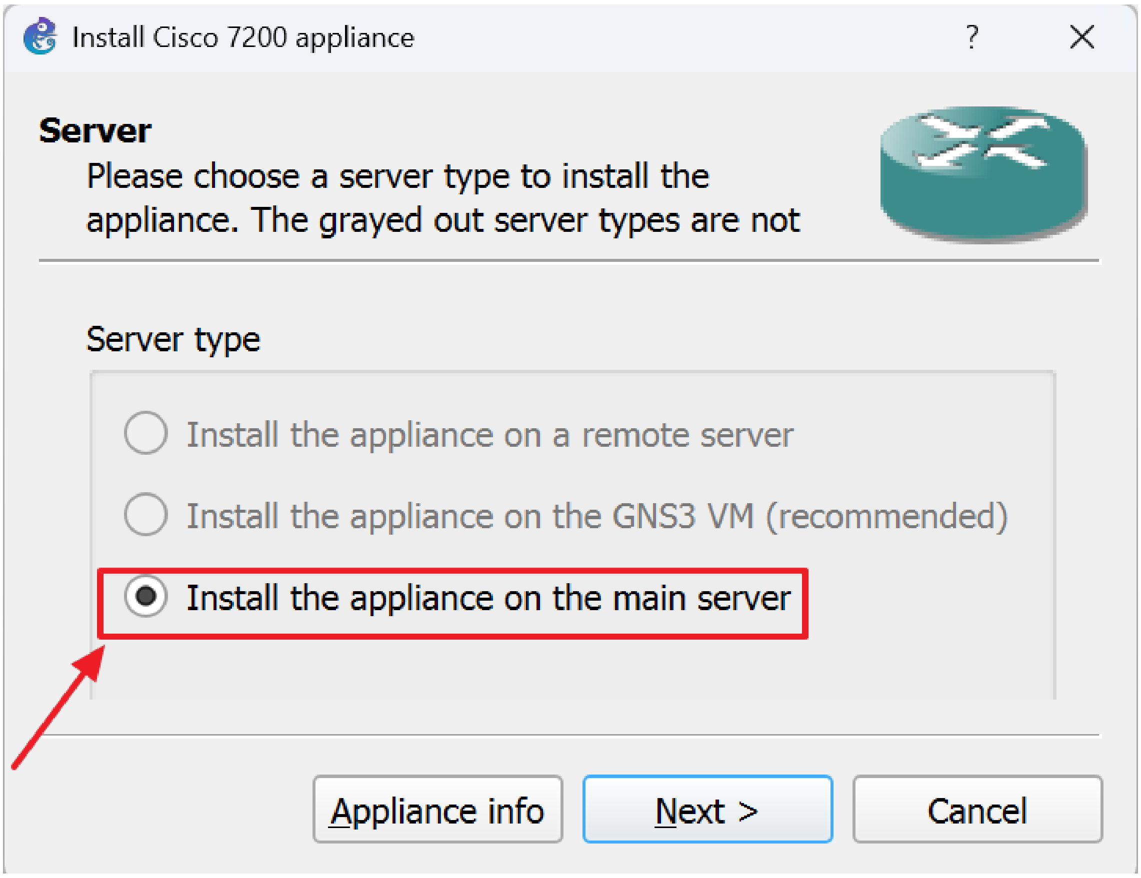


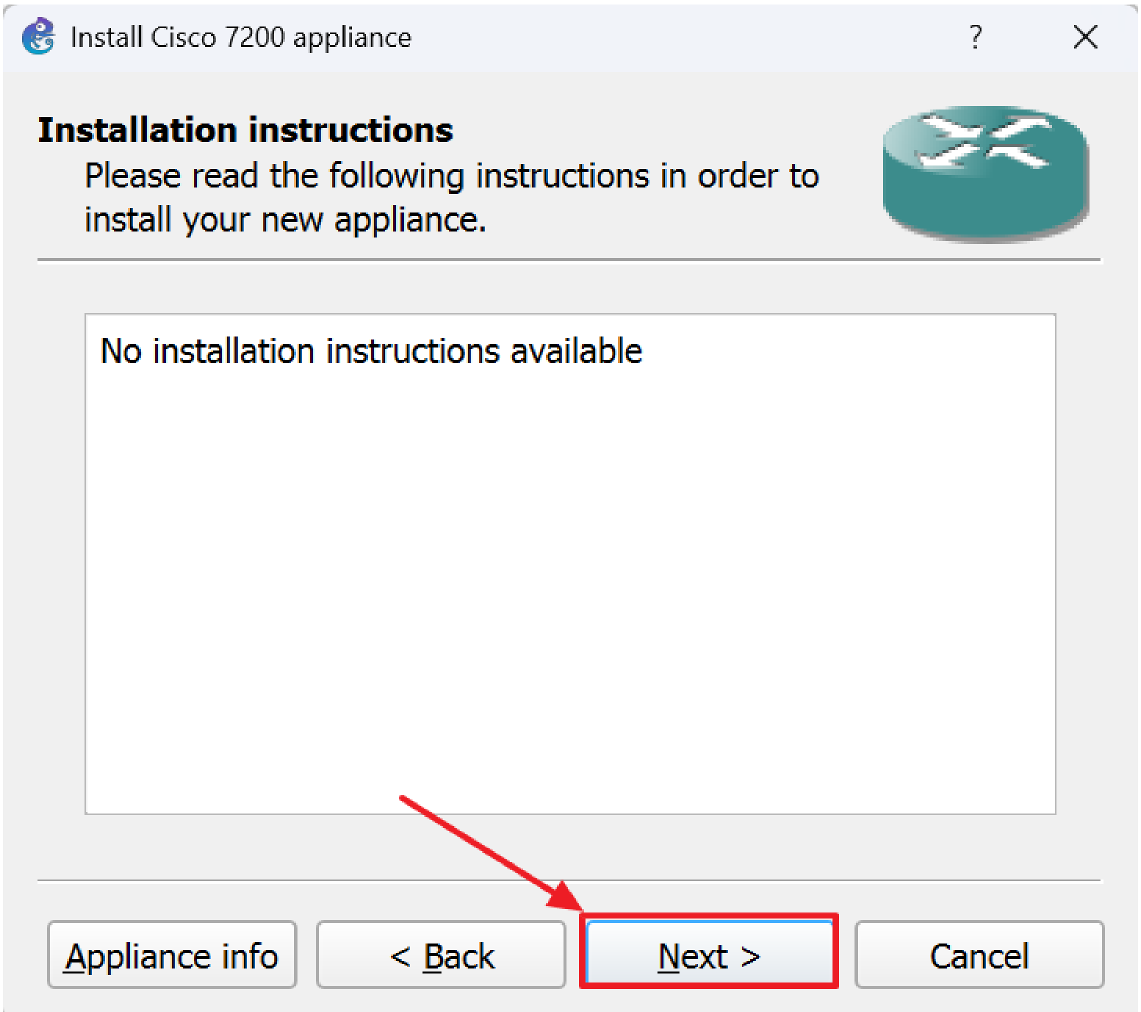
3. Importar Imágenes de Router a GNS3

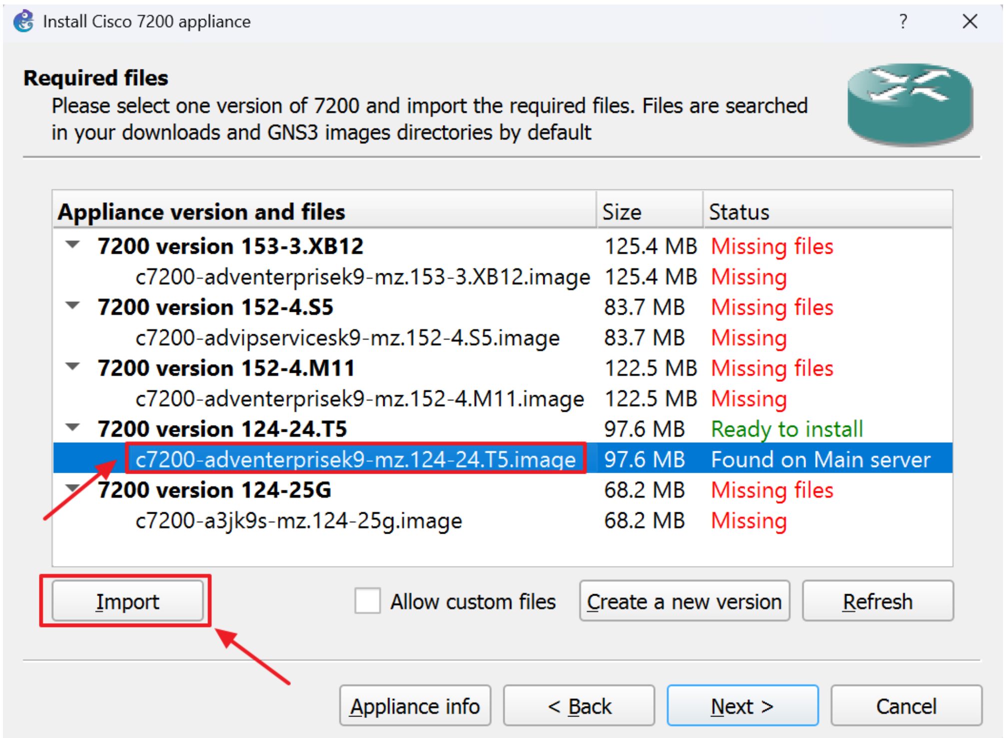


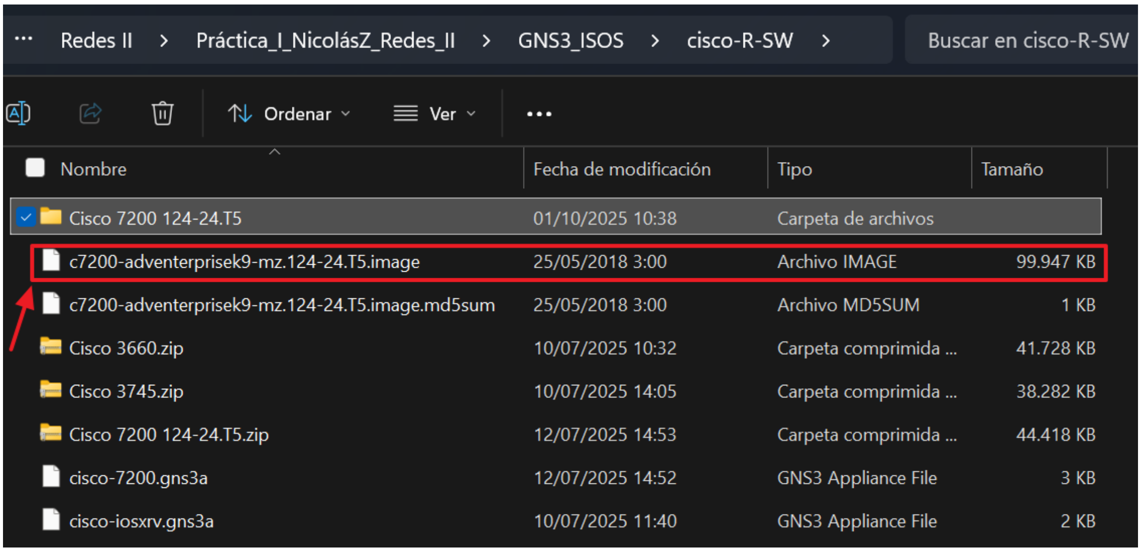



Import appliance
GNS3_ISOS › cisco-R-SW
Buscar en cisco-
Organizar
Nueva carpeta
OneDrive - Pers
Escritorio
Descargas
Nombre
Fecha de modificación
Cisco 7200 124-24.T5
01/10/2025 10:38
cisco-7200.gns3a
12/07/2025 14:52
cisco-iosxrv.gns3a
10/07/2025 11:40


Install Cisco 7200 appliance
Server
Please choose a server type to install the appliance. The grayed out server types are not
Server type
Install the appliance on a remote server
Install the appliance on the GNS3 VM (recommended)
Install the appliance on the main server
Appliance info
Next >
Cancel


Install Cisco 7200 appliance
Installation instructions
Please read the following instructions in order to install your new appliance.
No installation instructions available
Appliance info
< Back
Next >
Cancel


Install Cisco 7200 appliance
Required files
Please select one version of 7200 and import the required files. Files are searched in your downloads and GNS3 images directories by default
Appliance version and files
Size
Status
7200 version 153-3.XB12
125.4 MB
Missing files
c7200-adventerprisek9-mz.153-3.XB12.image
125.4 MB
Missing
7200 version 152-4.S5
83.7 MB
Missing files
c7200-advipservicesk9-mz.152-4.S5.image
83.7 MB
Missing
7200 version 152-4.M11
122.5 MB
Missing files
c7200-adventerprisek9-mz.152-4.M11.image
122.5 MB
Missing
7200 version 124-24.T5
97.6 MB
Ready to install
c7200-adventerprisek9-mz.124-24.T5.image
97.6 MB
Found on Main server
7200 version 124-25G
68.2 MB
Missing files
c7200-a3jk9s-mz.124-25g.image
68.2 MB
Missing
Import
Allow custom files
Create a new version
Refresh
Appliance info
< Back
Next >
Cancel



Redes II > Práctica_I_NicolásZ_Redes_II > GNS3_ISOS > cisco-R-SW >
Buscar en cisco-R-SW
Ordenar
Ver
Nombre
Fecha de modificación
Tipo
Tamaño
Cisco 7200 124-24.T5 01/10/2025 10:38 Carpeta de archivos
c7200-adventerprisek9-mz.124-24.T5.image 25/05/2018 3:00 Archivo IMAGE 99.947 KB
c7200-adventerprisek9-mz.124-24.T5.image.md5sum 25/05/2018 3:00 Archivo MD5SUM 1 KB
Cisco 3660.zip 10/07/2025 10:32 Carpeta comprimida ... 41.728 KB
Cisco 3745.zip 10/07/2025 14:05 Carpeta comprimida ... 38.282 KB
Cisco 7200 124-24.T5.zip 12/07/2025 14:53 Carpeta comprimida ... 44.418 KB
cisco-7200.gns3a 12/07/2025 14:52 GNS3 Appliance File 3 KB
cisco-iosxrv.gns3a 10/07/2025 11:40 GNS3 Appliance File 2 KB

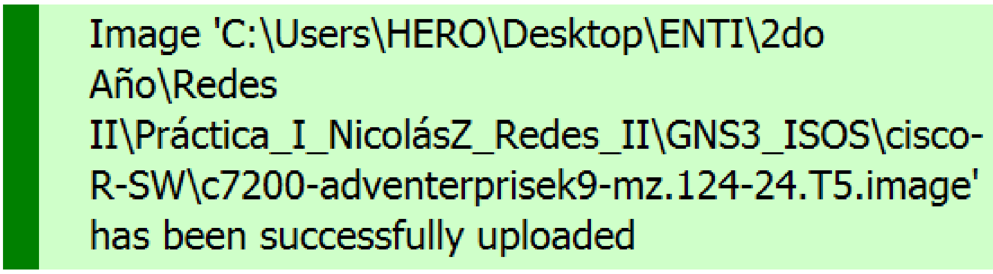
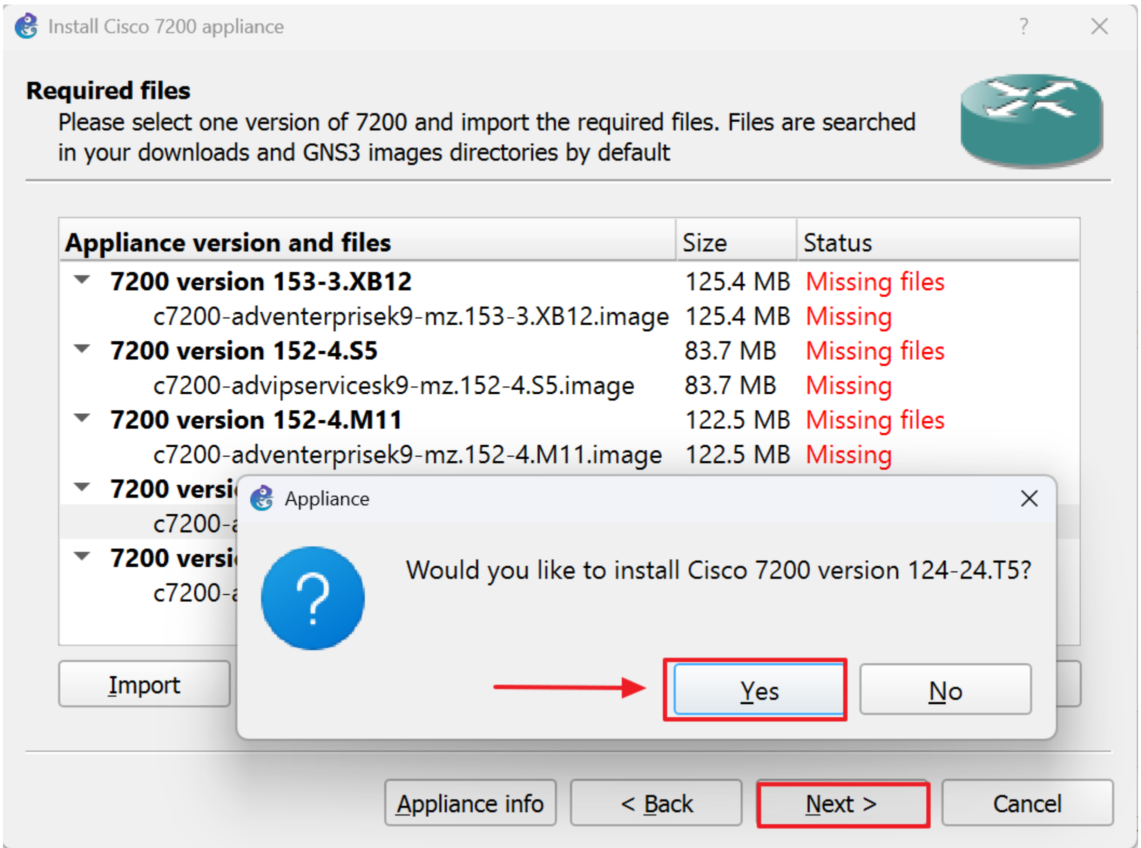
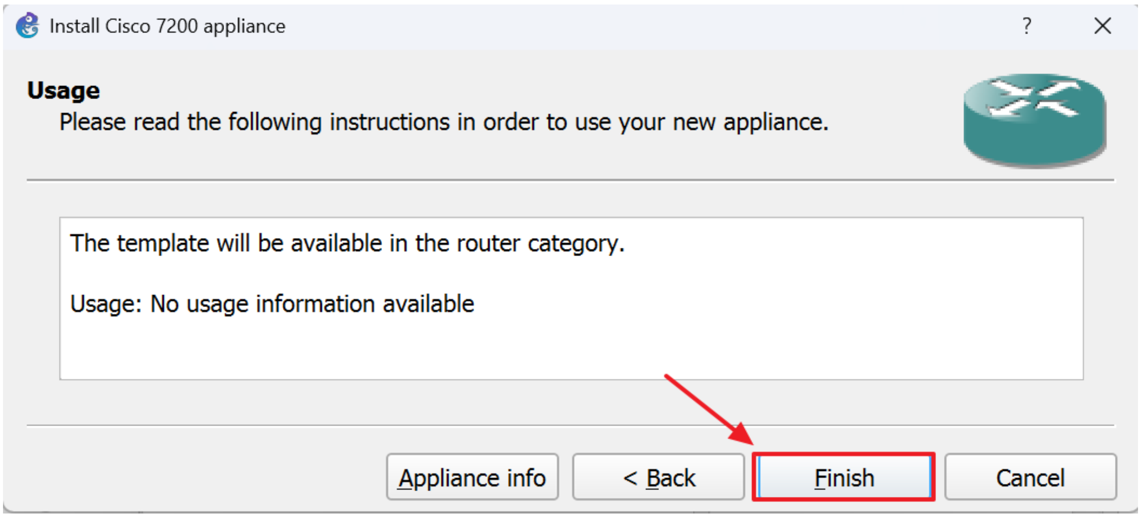
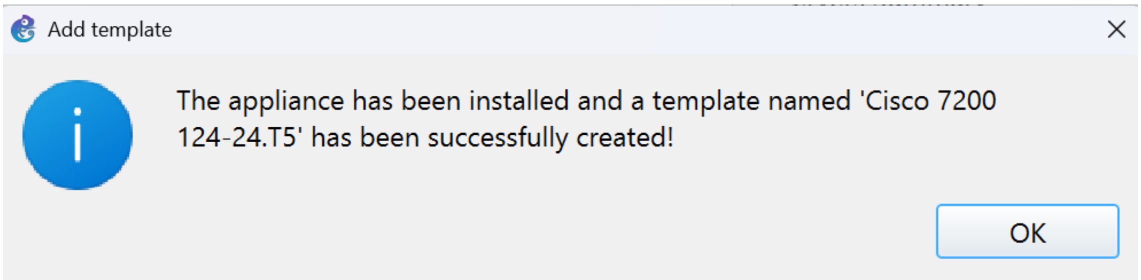


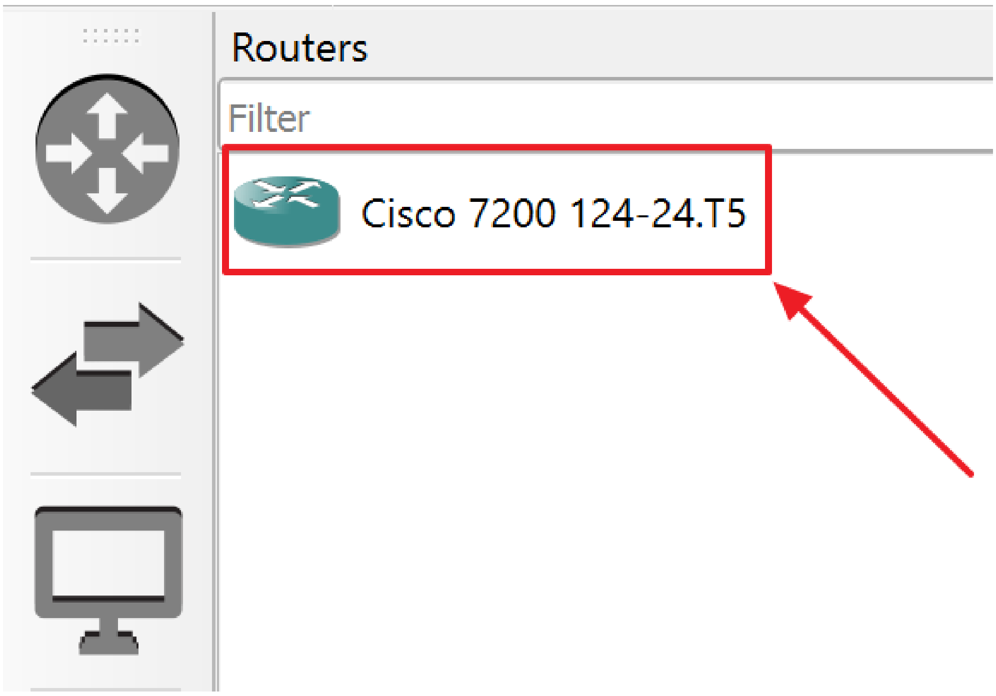
Image 'C:\Users\HERO\Desktop\ENTI\2do Año\Redes II\Práctica_I_NicolásZ_Redes_II\GNS3_ISOS\cisco-R-SW\c7200-adventerprisek9-mz.124-24.T5.image' has been successfully uploaded



Install Cisco 7200 appliance
Required files
Please select one version of 7200 and import the required files. Files are searched in your downloads and GNS3 images directories by default
Appliance version and files
Size
Status
7200 version 153-3.XB12 125.4 MB Missing files
c7200-adventerprisek9-mz.153-3.XB12.image 125.4 MB Missing
7200 version 152-4.S5 83.7 MB Missing files
c7200-advipservicesk9-mz.152-4.S5.image 83.7 MB Missing
7200 version 152-4.M11 122.5 MB Missing files
c7200-adventerprisek9-mz.152-4.M11.image 122.5 MB Missing
Import
Appliance
Would you like to install Cisco 7200 version 124-24.T5?
Yes
No
Appliance info
< Back
Next >
Cancel


Install Cisco 7200 appliance
Usage
Please read the following instructions in order to use your new appliance.
The template will be available in the router category.
Usage: No usage information available
Appliance info
< Back
Finish
Cancel


Add template
The appliance has been installed and a template named 'Cisco 7200 124-24.T5' has been successfully created!
OK


Routers
Filter
Cisco 7200 124-24.T5

4. Importar imágenes CISCO IOU en GNS3

Para importar los Switches virtuales de Cisco utilizaremos las imágenes del siguiente repositorio:

https://github.com/hegdepavankumar/Cisco-Images-for-GNS3-and-EVE-NG?tab=readme-ov-file

Como podemos observar, tenemos imágenes de Cisco capa 2 y capa 3.

Las imágenes de capa 3 vienen con funciones de Routing (OSPF, EIGRP, entre otras), y Switching VLANs, Spanning Tree, Trunks, etc). En cambio, las de capa 2 solo Switching.

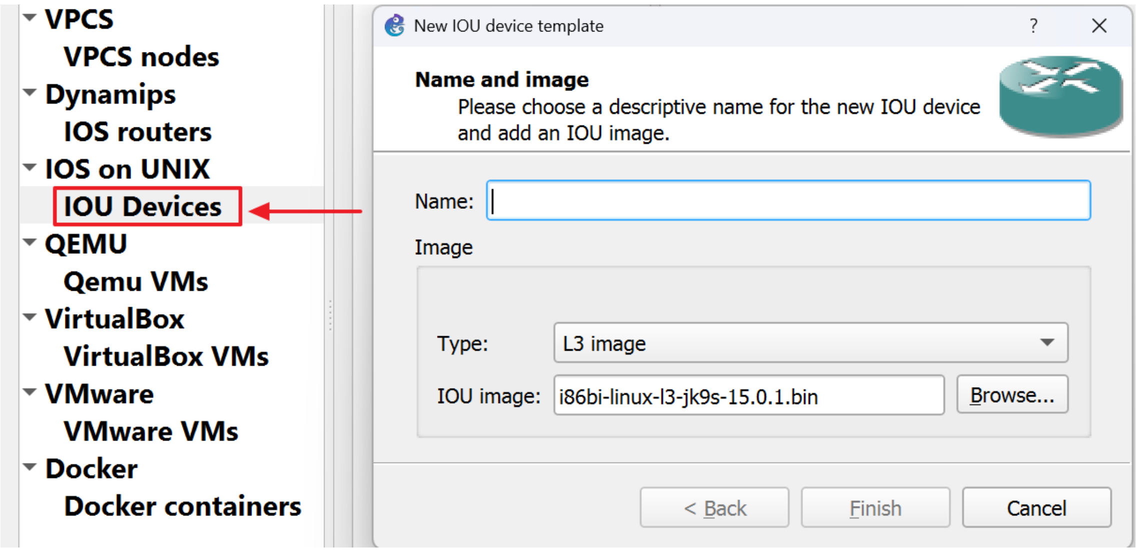
Importaremos a nuestro servidor GNS3 las siguientes imágenes:

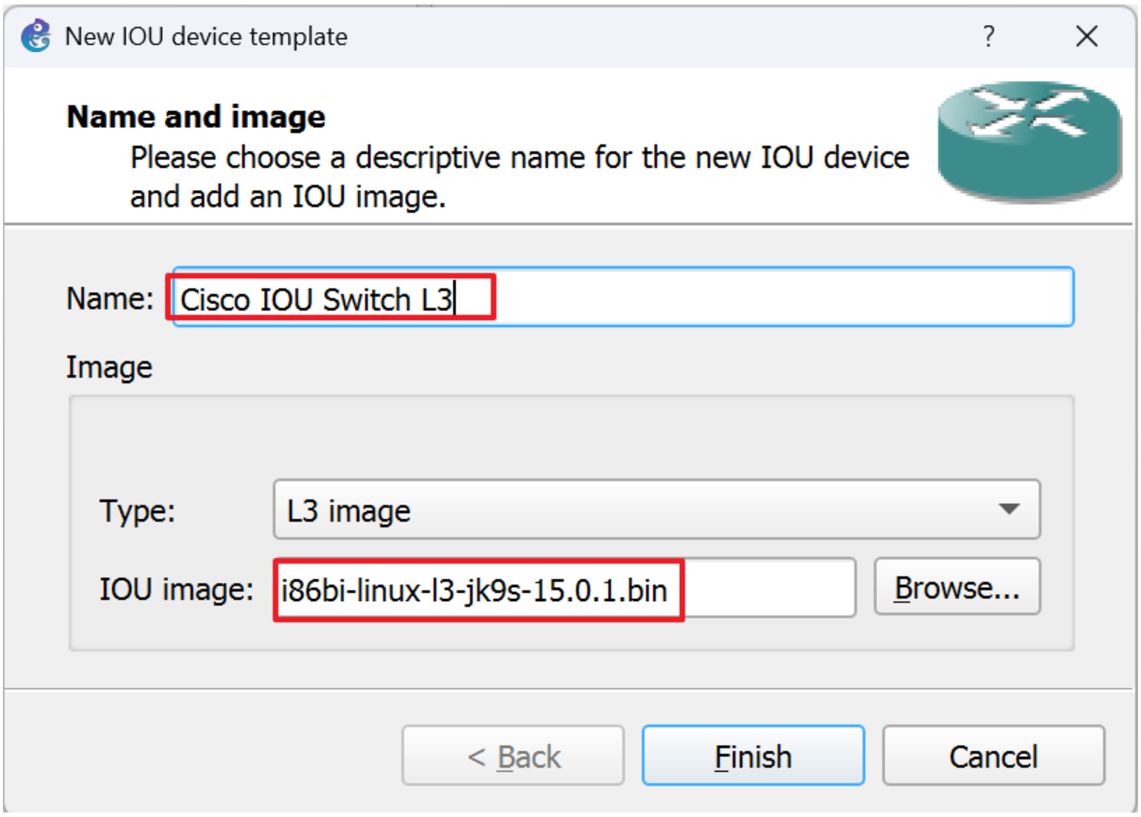
Cisco IOU Images For GNS3

SR	CISCO IOU NAME	DOWNLOAD	TYPE
1	i86bi-linux-l2-adventerprise-15.1b.bin	Download here	L2
2	i86bi-linux-l2-adventerprisek9-15.1a.bin	Download here	L2
3	i86bi-linux-l2-ipbasek9-15.1a.bin	Download here	L2
4	i86bi-linux-l2-ipbasek9-15.1b.bin	Download here	L2
5	i86bi-linux-l2-ipbasek9-15.1c.bin	Download here	L2
6	i86bi-linux-l2-ipbasek9-15.1d.bin	Download here	L2
7	i86bi-linux-l2-ipbasek9-15.1e.bin	Download here	L2
8	i86bi-linux-l2-ipbasek9-15.1f.bin	Download here	L2
9	i86bi-linux-l2-ipbasek9-15.1g.bin	Download here	L2
10	i86bi-linux-l2-upk9-12.2.bin	Download here	L2
11	i86bi-linux-l2-upk9-15.0a.bin	Download here	L2
12	i86bi-linux-l2-upk9-15.0b.bin	Download here	L2
13	i86bi-linux-l3-jk9s-15.0.1.bin	Download here	L3
14	i86bi-linux-l3-p-15.0a.bin	Download here	L3
15	i86bi-linux-l3-p-15.0b.bin	Download here	L3
16	i86bi-linux-l3-tpgen-adventerprisek9-12.4.bin	Download here	L3

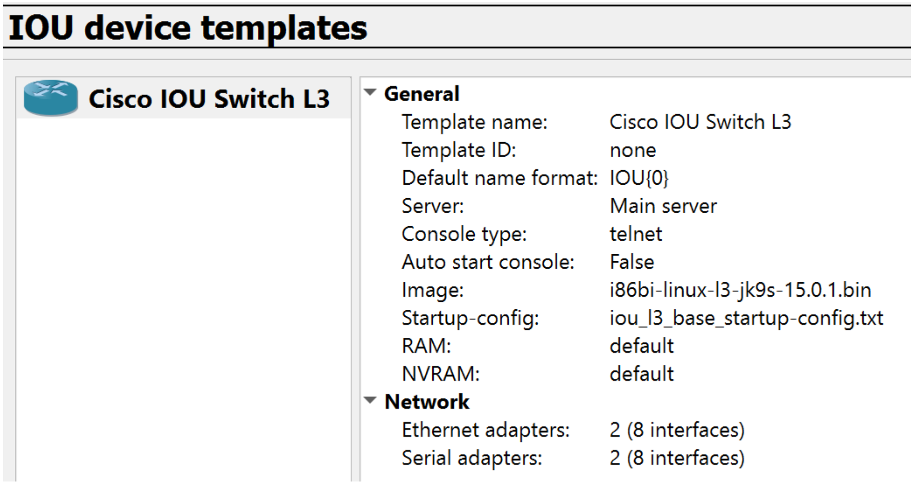
Vamos a edit >> preferences > IOU Devices.

Nos pregunta que tipo de imagen es, lo pondremos como L3 (Capa3).

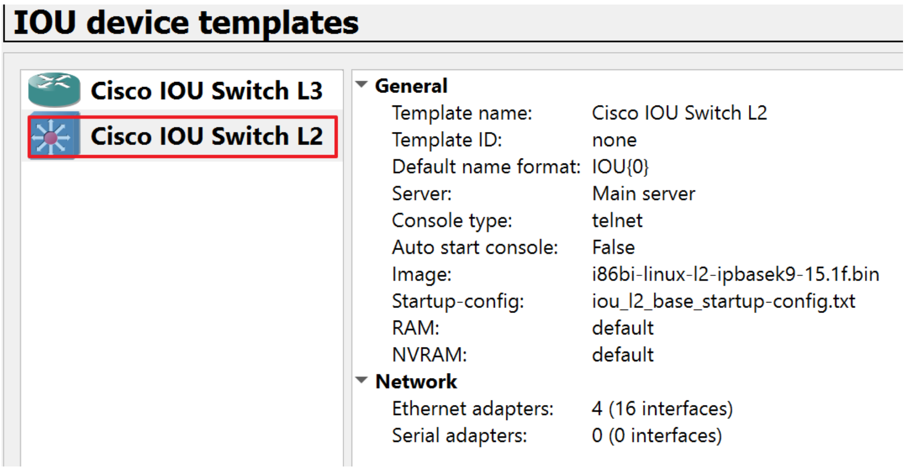




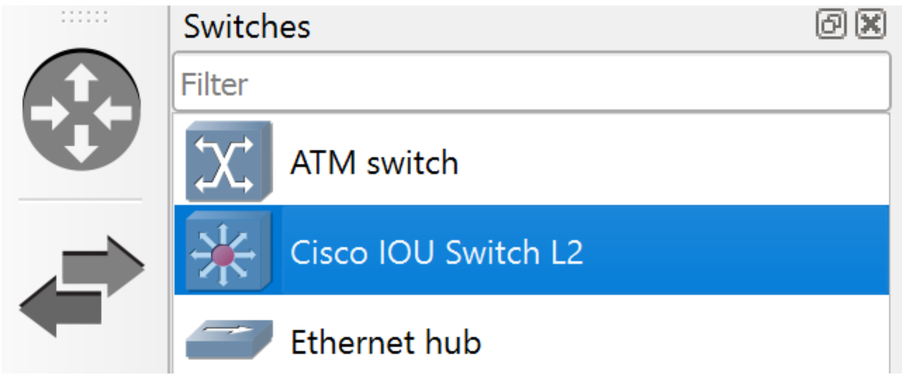
Ya tenemos incorporado un Dispositivo CAPA3.



Ahora seguimos los siguientes pasos con la otra imagen seleccionada anteriormente y la importamos a GNS3.



Ya tenemos el Switch incorporado:

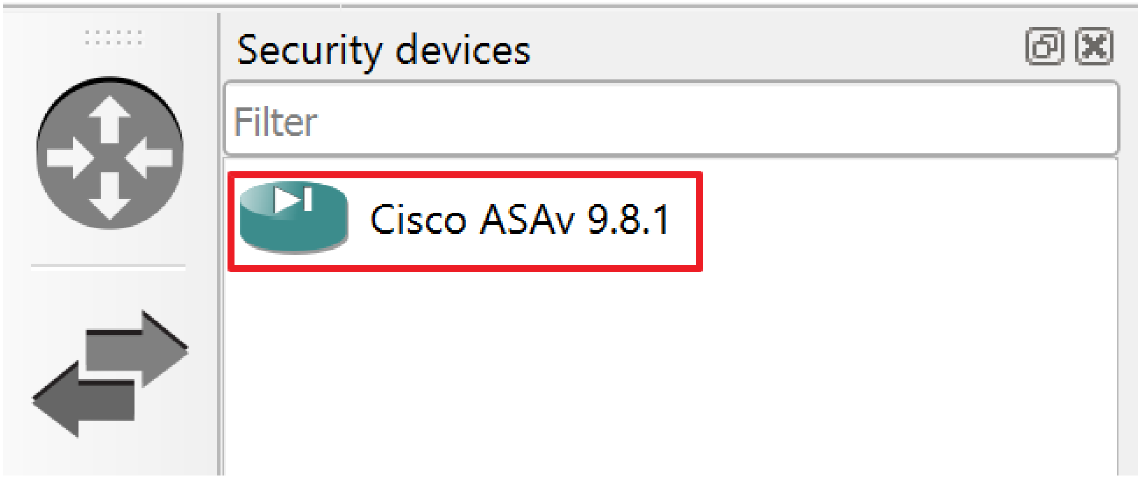


Ahora incorporaremos un firewall Cisco ASA.

Cisco ASA Firewall Images For GNS3

SR	CISCO IOS NAME	DOWNLOAD	VERSION
1	asav981.qcow2	Download here	ASAv 9.8.3
2	asav991.qcow2	Download here	ASAv 9.9.1
3	asav992-32.qcow2	Download here	ASAv 9.9.2
4	asav981.qcow2	Download here	ASAv 9.8.1
5	asa842-initrd.gz	Download here	ASA 8.4.2
6	asa-915-k8.qcow2	Download here	ASA 9.1.5

Lo importamos y ya tenemos un firewall listo para usar.



Te conectas via Putty al Main Server.

>> sudo su

>> sudo apt-get update

>> sudo apt-get install python3

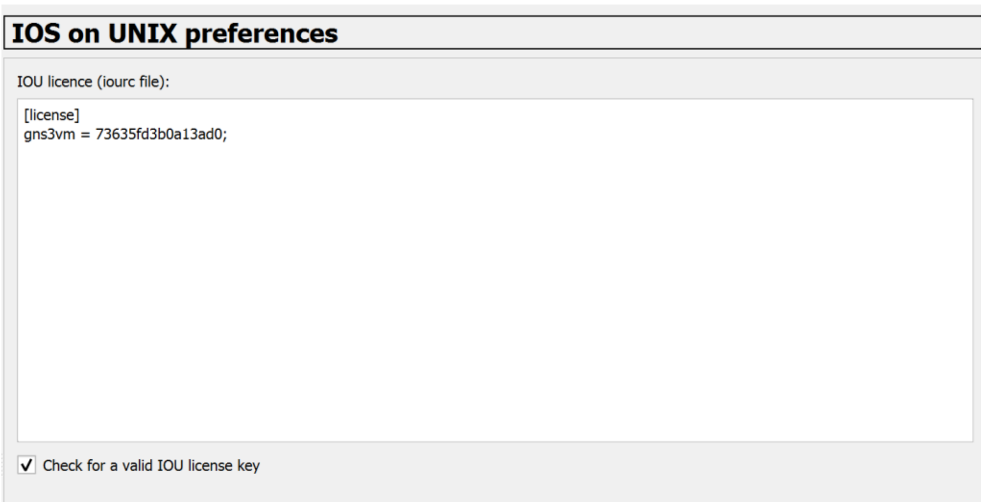
>> wget http://www.ipvanquish.com/download/CiscoIOUKeygen3f.py

>> python3 CiscoIOUKeygen3f.py

Nos sacará la siguiente info:

```
gns3@gns3vm:~$ ls -l
total 8
-rw-rw-r-- 1 gns3 gns3 1201 Nov 12  2016 CiscoIOUKeygen3f.py
drwxr-xr-x 5 gns3 gns3 4096 Apr 21 12:43 GNS3
gns3@gns3vm:~$ python3 CiscoIOUKeygen3f.py
*********************************************************************
Cisco IOU License Generator - Kal 2011, python port of 2006 C version
hostid=00000000, hostname=gns3vm, ioukey=25e

Add the following text to ~/.iourc:
[license]
gns3vm = 73635fd3b0a13ad0;

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
Already copy to the file iourc.txt

You can disable the phone home feature with something like:
 echo '127.0.0.127 xml.cisco.com' >> /etc/hosts
```

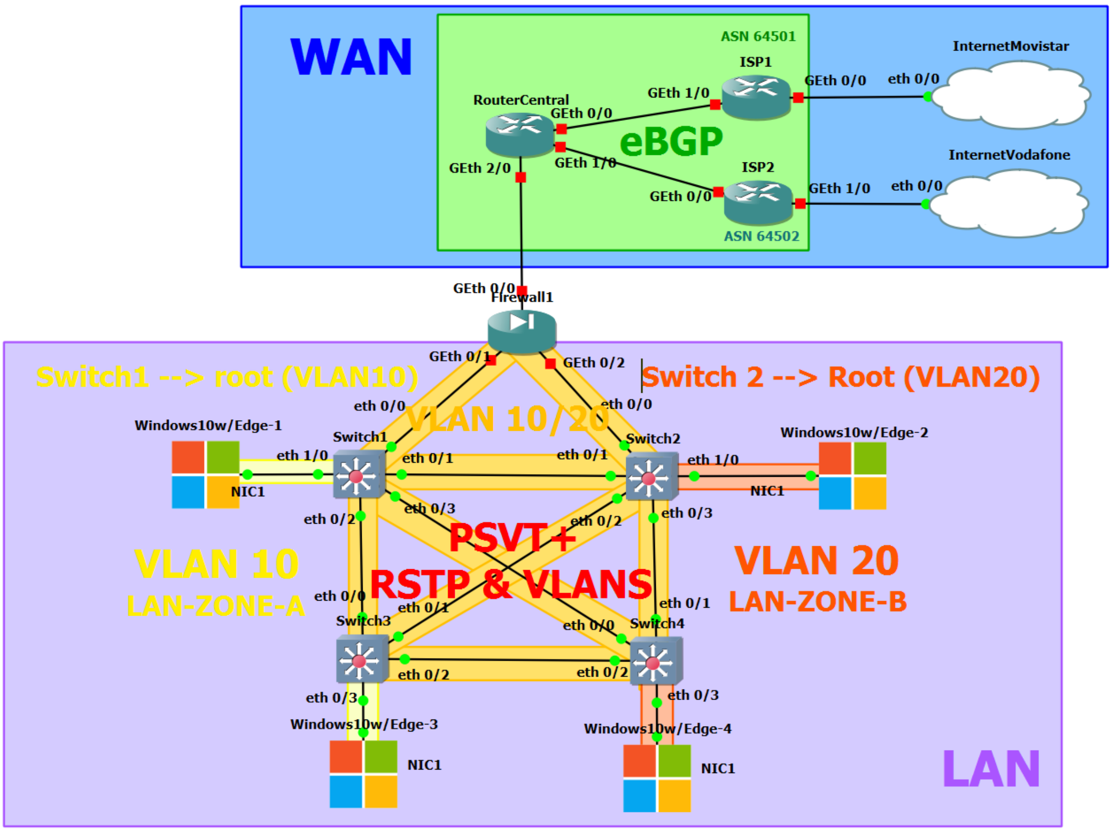
Ahora nos vamos a GNS3 >> Edit >> Preferences >> IOS on UNIX preferences y copiamos la información de licencia.



A la hora de configurar los puertos podemos encontrar diferentes tipos de puertos:

Módulo	**Tipo de interfaz**	**Medio físico**	**Velocidad aprox.**	**Uso típico**
C7200-IO-{FE \ 2FE}	FastEthernet (1 puerto) Double FastEthernet (2 puertos)	Cobre (UTP/RJ-45)	100 Mbps c/p	LAN básica o conexión de gestión
C7200-IO-GE-E GE = GigabitEthernet	GigabitEthernet (1 puerto)	Cobre (UTP/RJ-45)	1 Gbps	LAN moderna / alta velocidad
PA-A1 A1 = ATM Asynchronous Transfer Mode)	Port Adapter-ATM OC-3 \ OC-12	Fibra óptica	155 Mbps (OC-3) 622 Mbps (OC-12)	Enlaces WAN ATM
PA-{FE / 2FE}-TX TX = Cable de cobre trenzado	Port Adapter - FastEthernet (1 puerto)	Cobre (UTP/RJ-45)	100 Mbps	Conexión LAN simple
PA-GE	Port Adapter - GigabitEthernet (1 puerto)	Cobre (UTP/RJ-45)	1 Gbps	LAN / WAN moderna, ideal para eBGP
PA-{4T+ - 8T}	Serial (4 puertos)	Serial (DB-60 / Smart Serial)	Hasta 8 Mbps por enlace	Enlaces WAN (PPP, Frame Relay)
PA-{4E / 8E}	Ethernet (4 puertos / 8 puertos).	Cobre (UTP/RJ-45)	10 Mbps	LAN antigua / laboratorio básico
PA-POS-OC3 POS = Packet over SONET	Packet over SONET (1 puerto)	**Fibra óptica**	155 Mbps (OC-3)	Enlace WAN óptico (POS)

Una vez importadas las imágenes tenemos la siguiente red:



5. Configurar RSTP (Rapid Spanning Tree Protocol)

Procedemos a configurar la siguiente red:

```
SWITCH1 -- SWITCH2
   |    \     /   |
   |      \  /    |
   |      /  \    |
   |    /     \   |
SWITCH3----SWITCH4
```

-Switch1:

Switch1 (eth 0/1) --> Switch2 (eth 0/1)

Switch1 (eth 0/2) --> Switch3 (eth 0/0)

Switch1 (eth 0/3) --> Switch4 (eth 0/0)

Switch1 (eth 1/0) --> Windows10w/Edge-1 (NIC1)

-Switch2:

Switch2 (eth 0/1) --> Switch1 (eth 0/1)

Switch2 (eth 0/2) --> Switch3 (eth 0/1)

Switch2 (eth 0/3) --> Switch4 (eth 0/1)

Switch2 (eth 1/0) --> Windows10w/Edge-2 (NIC1)

-Switch3:

Switch3 (eth 0/0) --> Switch1 (eth 0/2)

Switch3 (eth 0/1) --> Switch2 (eth 0/2)

Switch3 (eth 0/2) --> Switch4 (eth 0/2)

Switch3 (eth 0/3) --> Windows10w/Edge-3 (NIC1)

-Switch4:

Switch4 (eth 0/0) --> Switch1 (eth 0/3)

Switch4 (eth 0/1) --> Switch2 (eth 0/3)

Switch4 (eth 0/2) --> Switch3 (eth 0/2)

-Windows10w/Edge-4 (NIC1) --> Switch4 (eth 0/3)

'---Configuar PVST+---'

1-- Borrar configuración y reiniciar switch

>> enable

>> erase startup-config

>> reload

2-- Cambiar nombre

>> enable

>> config terminal

>> hostname Switch2

>> exit

>> copy running-config startup-config

3-- Activar PVST+ (RSTP con soporte para VLANs).

>> enable

>> configure terminal

>> spanning-tree mode rapid-pvst

>> end

>> copy running-config startup-config

4-- Buscar configuracion:

>> enable

>> show running-config | include spanning-tree

'---Configuguar básica de interfaces---'

1-- Crear VLANs

>> enable

>> config terminal

>> vlan 10

>> name LAN-ZONE-{A, B}

>> copy running-config startup-config

2-- Consultar VLANs

>> show vlan brief

3-- Configurar los puertos troncales (SWITCH1 y SWITCH2).

Switch1 (eth0/0, eth0/1, eth0/2, eth0/3):

>> enable

>> conf t

>> interface eth0/{0,1,2,3}

>> switchport trunk encapsulation dot1q

>> switchport mode trunk

>> switchport trunk allowed vlan 10,20

>> exit

>> show interfaces trunk

>> copy running-config startup-config

-- Después configurar el puerto eth0/2 del Switch1:

>> enable

>> conf t

>> interface eth1/0

>> switchport mode access

>> switchport access vlan 10

>> spanning-tree portfast

>> spanning-tree bpduguard enable

>> exit

>> show vlan brief

>> show spanning-tree interface eth1/0

>> copy running-config startup-config

Switch2 (eth0/0, eth0/1, eth0/2, eth0/3):

>> enable

>> conf t

>> interface eth0/{0,1,2,3}

>> switchport trunk encapsulation dot1q

>> switchport mode trunk

>> switchport trunk allowed vlan 10,20

>> exit

>> show interfaces trunk

>> copy running-config startup-config

-- Después configurar el puerto eth1/0 del Switch2:

>> enable

>> conf t

>> interface eth1/0

>> switchport mode access

>> switchport access vlan 20

>> spanning-tree portfast

>> spanning-tree bpduguard enable

>> exit

>> show vlan brief

>> show spanning-tree interface eth1/0

>> copy running-config startup-config

3.1--Configurar puertos troncales (SWITCH3 y SWITCH4).

Switch 3 (eth0/0, eth0/1, eth0/2):

>> enable

>> conf t

>> interface eth0/{0,1,2}

>> switchport trunk encapsulation dot1q

>> switchport mode trunk

>> switchport trunk allowed vlan 10,20

>> exit

>> show interfaces trunk

>> copy running-config startup-config

-- Después configurar el puerto eth0/3 del Switch3:

>> enable

>> conf t

>> interface eth0/3

>> switchport mode access

>> switchport access vlan 10

>> spanning-tree portfast

>> spanning-tree bpduguard enable

>> exit

>> show vlan brief

>> show spanning-tree interface eth0/3

>> copy running-config startup-config

Switch 4 (eth0/0, eth0/1, eth0/2):

>> enable

>> conf t

>> interface eth0/{0,1,2}

>> switchport trunk encapsulation dot1q

>> switchport mode trunk

>> switchport trunk allowed vlan 10,20

>> exit

>> show interfaces trunk

>> copy running-config startup-config

-- Después configurar el puerto eth0/3 del Switch3:

>> enable

>> conf t

>> interface eth0/3

>> switchport mode access

>> switchport access vlan 20

>> spanning-tree portfast

>> spanning-tree bpduguard enable

>> exit

>> show vlan brief

>> show spanning-tree interface eth0/3

>> copy running-config startup-config

4-- Establecer ROOT para VLAN 10 Y VLAN 20.

Switch 1:

PARA VLAN 10:

>> enable

>> conf t

>> spanning-tree vlan 10 priority 32768

>> exit

>> copy running-config startup-config

PARA VLAN 20:

>> enable

>> conf t

>> spanning-tree vlan 20 priority 4096

>> exit

>> copy running-config startup-config

>> show spanning-tree vlan 10

>> show spanning-tree vlan 20

Switch 2:

PARA VLAN 20:

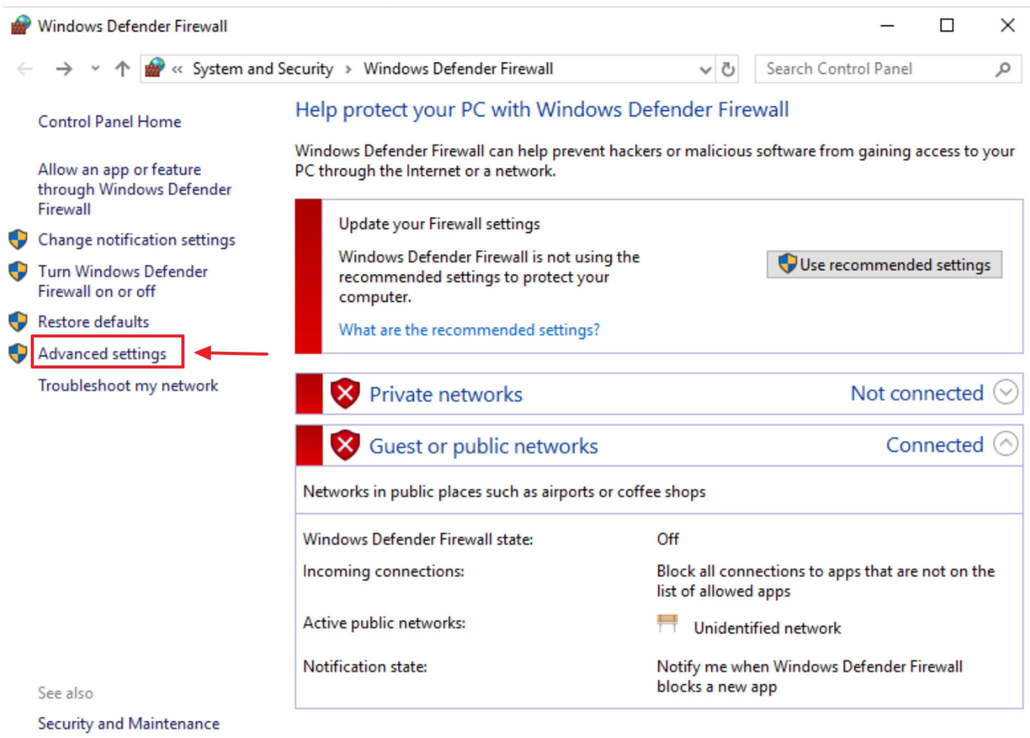
>> enable

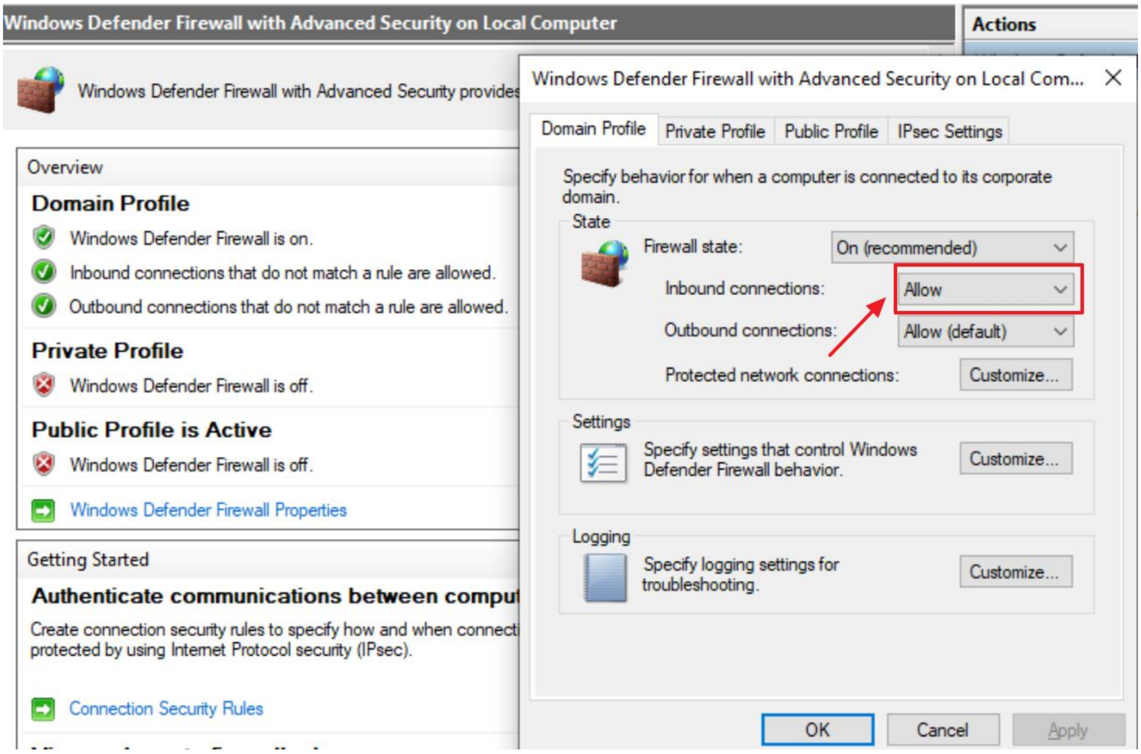
```
>> conf t
>> spanning-tree vlan 20 priority 4096
>> exit
>> copy running-config startup-config
>> show spanning-tree vlan 10
>> show spanning-tree vlan 20
```

5-- Asignar IPs a los clientes:

	VLAN	IP	Máscara	Puerta de enlace
Edge1	10	192.168.10.100	255.255.255.0	192.168.10.1
Edge3	10	192.168.10.103	255.255.255.0	192.168.10.1
Edge2	20	192.168.20.102	255.255.255.0	192.168.20.1
Edge4	20	192.168.20.104	255.255.255.0	192.168.20.1

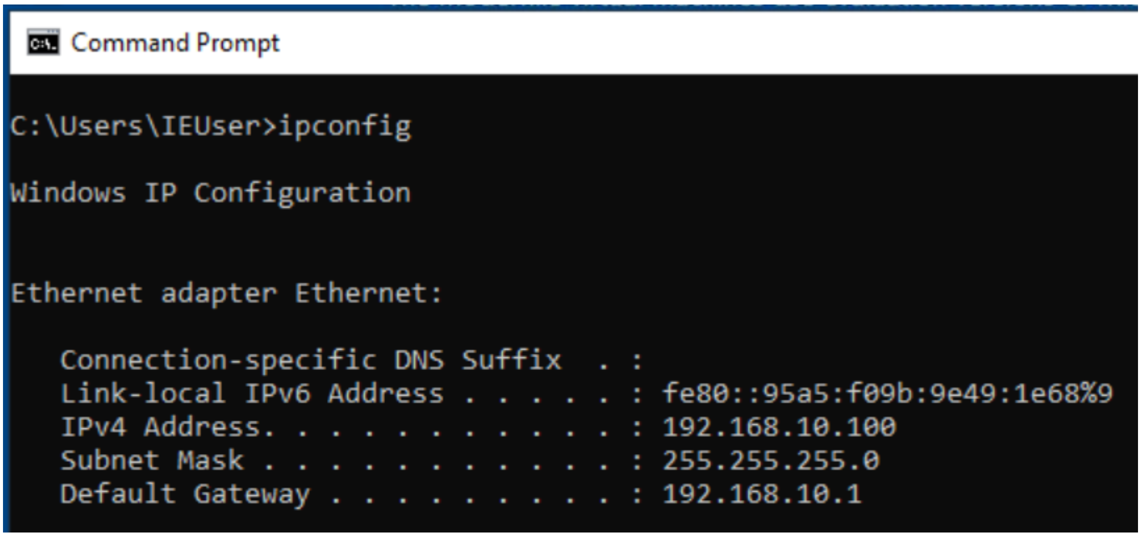
Desactivamos Windows Defender:





CONFIGURACIÓN PCs / PRUEBAS DE CONECTIVIDAD

WIN10-EDGE1:



WIN10-EDGE2:

```
Command Prompt

C:\Users\IEUser>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::396c:7bf7:7266:9528%7
   IPv4 Address. . . . . . . . . . . : 192.168.20.102
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.20.1
```

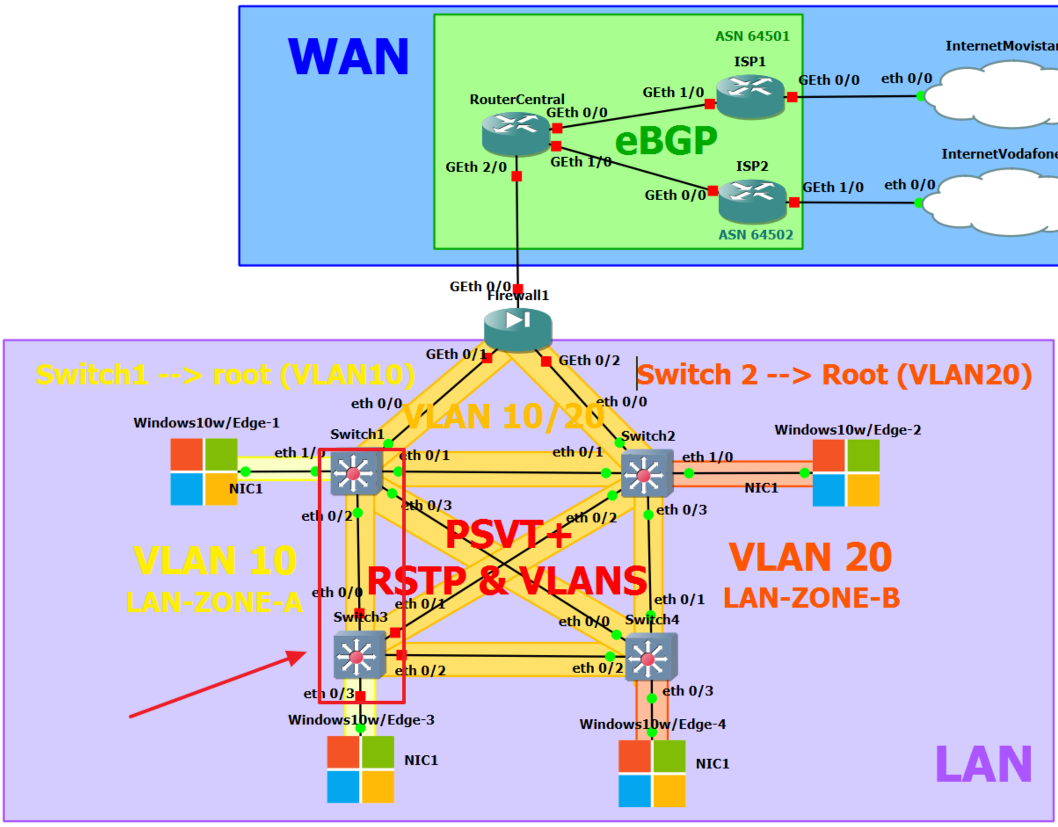
WIN10-EDGE3:

```
C:\Windows\System32\cmd.exe

C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::183c:38a7:c67d:1573%5
   IPv4 Address. . . . . . . . . . . : 192.168.10.103
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.10.1
```

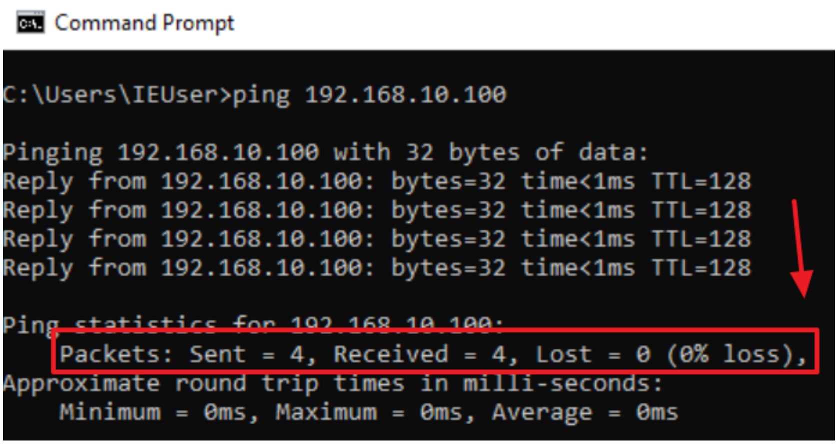
WIN10-EDGE4:

```
Command Prompt

C:\Users\IEUser>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fcb5:1a53:3ed1:db63%7
   IPv4 Address. . . . . . . . . . . : 192.168.20.104
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.20.1
```

Pings entre PCs (Con todos los enlaces)



WIN10-EDGE1 (VLAN10) → WIN10-EDGE1 (VLAN10)



WIN10-EDGE1 (VLAN10) → WIN10-EDGE3 (VLAN10)

Command Prompt

```
C:\Users\IEUser>ping 192.168.10.103

Pinging 192.168.10.103 with 32 bytes of data:
Reply from 192.168.10.103: bytes=32 time=2ms TTL=128
Reply from 192.168.10.103: bytes=32 time=1ms TTL=128
Reply from 192.168.10.103: bytes=32 time=1ms TTL=128
Reply from 192.168.10.103: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

WIN10-EDGE1 (VLAN10) → WIN10-EDGE2 (VLAN20)

Command Prompt

```
C:\Users\IEUser>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

WIN10-EDGE1 (VLAN10) → WIN10-EDGE4 (VLAN20)

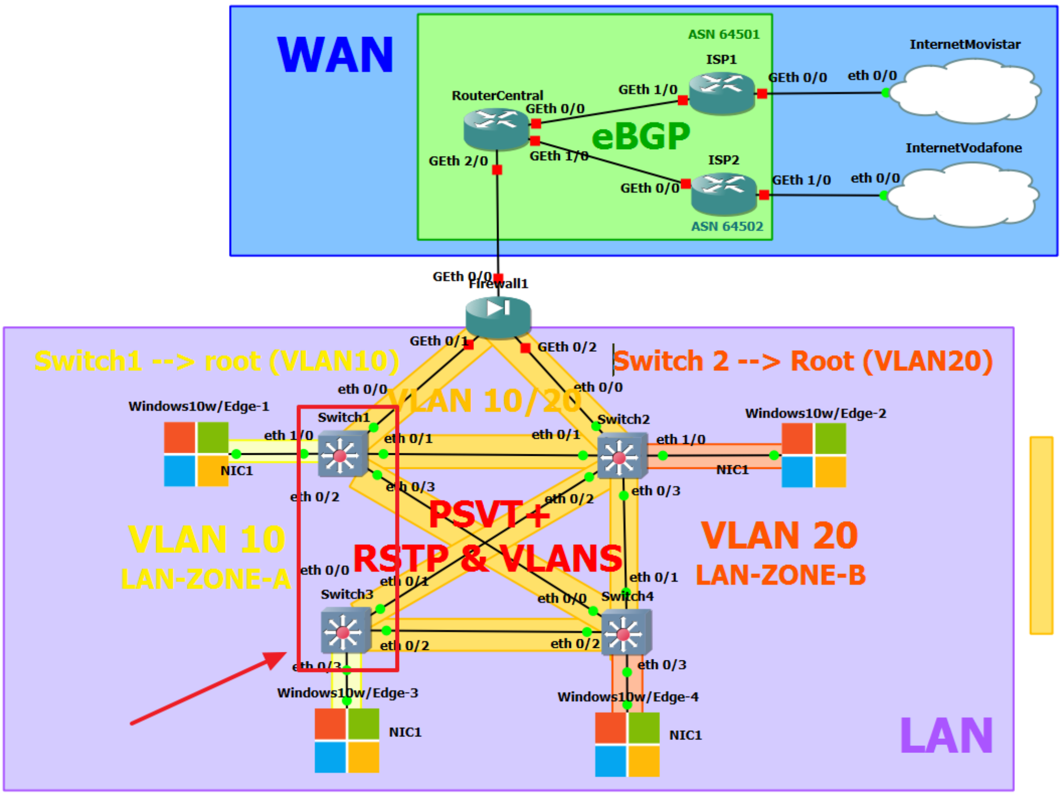
Command Prompt

```
C:\Users\IEUser>ping 192.168.20.104

Pinging 192.168.20.104 with 32 bytes of data:
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.

Ping statistics for 192.168.20.104:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Pruebas de conectividad eliminando un enlace en la malla de Switches (PSTV+).

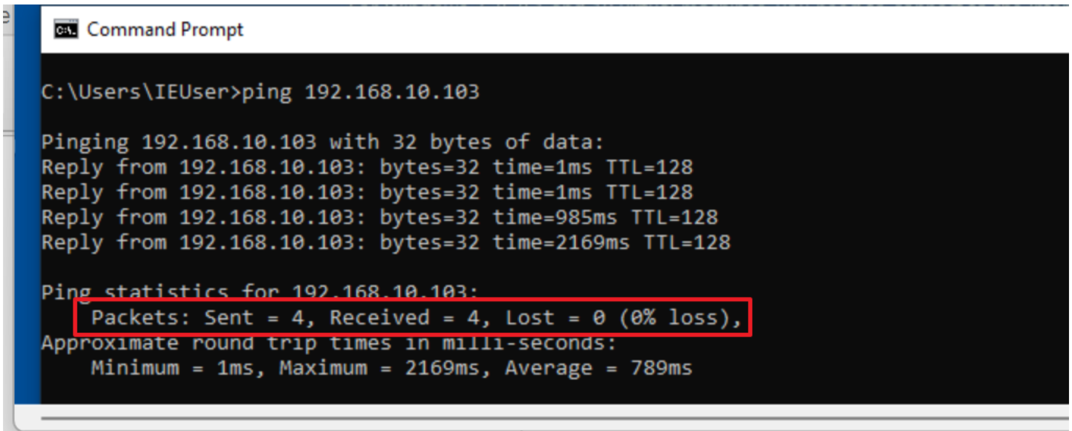
Procedemos a eliminar enlace de Switch1 al Switch3 (VLAN10). En un caso normal debería fallarnos el ping…sin embargo, tenemos habilitado PSTV+, lo que nos da tolerancia al fallo de un enlace en la malla.

- Eliminación de enlace de Switch1 ←→ Switch3.

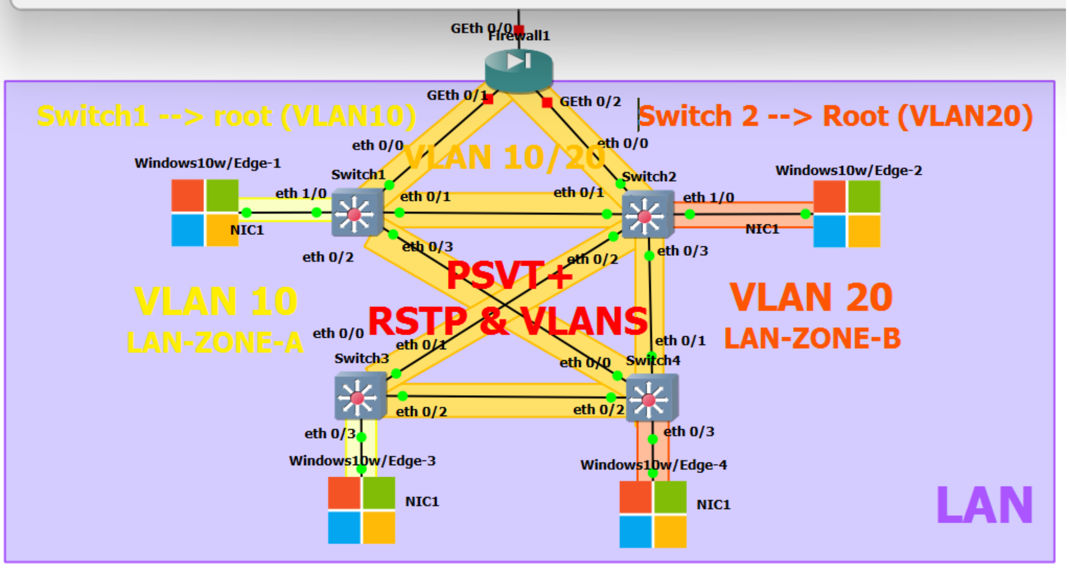


Pings entre PCs sin enlace de Switch1 a Switch3.

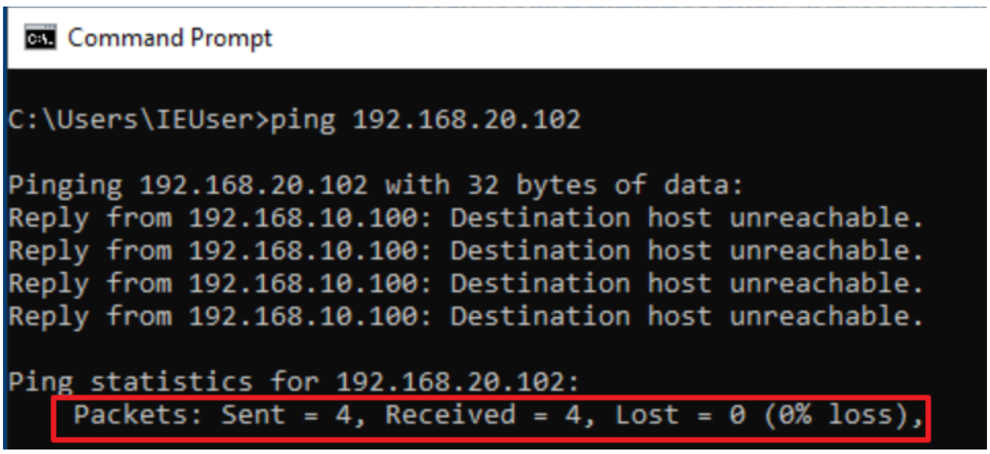
WIN10-EDGE1 (VLAN10) → WIN10-EDGE3 (VLAN10)



```
Command Prompt
C:\Users\IEUser>ping 192.168.10.103

Pinging 192.168.10.103 with 32 bytes of data:
Reply from 192.168.10.103: bytes=32 time=1ms TTL=128
Reply from 192.168.10.103: bytes=32 time=1ms TTL=128
Reply from 192.168.10.103: bytes=32 time=985ms TTL=128
Reply from 192.168.10.103: bytes=32 time=2169ms TTL=128

Ping statistics for 192.168.10.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2169ms, Average = 789ms
```



WIN10-EDGE1 (VLAN10) → WIN10-EDGE2 (VLAN20)



```
Command Prompt
C:\Users\IEUser>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.
Reply from 192.168.10.100: Destination host unreachable.

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Bloqueo de puertos a través de RSTP (Evitar tormenta de Broadcast).

-Switch1:

En SWITCH1 VLAN 10 todos los puertos configurados están como forwarding, ya que SWITCH1 es el ROOT BRIDGE de la VLAN 10.

```
Switch1#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    32778
             Address     aabb.cc00.0100
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     aabb.cc00.0100
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg FWD 100       128.33   Shr
```

En SWITCH1 VLAN 20 todos los puertos configurados están como forwarding excepto el Eth0/1 que va hacia Switch2 (ROOT BRIDGE VLAN 20).

```
Switch1#show spanning-tree vlan 20

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     aabb.cc00.0200
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     aabb.cc00.0100
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
```

-Switch2:

En SWITCH2 VLAN 10 todos los puertos configurados están como forwarding, excepto el Eth0/1 que va hacia Switch1 (ROOT BRIDGE VLAN 10).

```
Switch2#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    32778
             Address     aabb.cc00.0100
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     aabb.cc00.0200
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
```

En SWITCH2 VLAN 20 todos los puertos configurados están como forwarding, ya que SWITCH2 es el ROOT BRIDGE de la VLAN20.

```
Switch2#show spanning-tree vlan 20

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     aabb.cc00.0200
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4116   (priority 4096 sys-id-ext 20)
             Address     aabb.cc00.0200
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 100       128.1    Shr
Et0/1               Desg FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr
Et1/0               Desg FWD 100       128.33   Shr Edge
```

-Switch3:

En SWITCH3 VLAN 10 vemos que el Eth0/0 es root, que enlaza a SWITCH1 (ROOT BRIDGE VLAN 10).

Observamos que Eth0/1 está en Alternate/BLK. El SWITCH3 automáticamente bloquea este puerto para evitar tormenta de Broadcast.

```
Switch3>Show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    32778
             Address     aabb.cc00.0100
             Cost        100
             Port        1 (Ethernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Root FWD 100       128.1    Shr
Et0/1               Altn BLK 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr Edge
```

En SWITCH3 VLAN 20 ocurre lo mismo que en la anterior. Eth0/1 es un enlace troncal ROOT (ENLAZA AL ROOT BRIDGE de la VLAN 20).

Observamos que Eth0/0 está en Alternate/BLK. El SWITCH3 automáticamente bloquea este puerto para evitar tormenta de Broadcast.

```
Switch3>show spanning-tree vlan 20

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     aabb.cc00.0200
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     aabb.cc00.0300
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Altn BLK 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Desg FWD 100       128.3    Shr
```

-Switch4:

En SWITCH4 VLAN 10 vemos que el Eth0/0 es root, que enlaza a SWITCH1 (ROOT BRIDGE VLAN 10).

Observamos que Eth0/1 y Eth0/2 están en Alternate/BLK. El SWITCH4 automáticamente bloquea estos puertos para evitar tormentas de Broadcast.

```
Switch4#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    32778
             Address     aabb.cc00.0100
             Cost        100
             Port        1 (Ethernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     aabb.cc00.0400
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Root FWD 100       128.1    Shr
Et0/1               Altn BLK 100       128.2    Shr
Et0/2               Altn BLK 100       128.3    Shr
```

En SWITCH4 VLAN 20 ocurre lo mismo que en la anterior. Eth0/1 es un enlace troncal ROOT (ENLAZA AL ROOT BRIDGE de la VLAN 20).

Observamos que Eth0/0 y Eth0/3 están en Alternate/BLK. El SWITCH4 automáticamente bloquea estos puertos para evitar tormenta de Broadcast.

```
Switch4#show spanning-tree vlan 20

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     aabb.cc00.0200
             Cost        100
             Port        2 (Ethernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
             Address     aabb.cc00.0400
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Altn BLK 100       128.1    Shr
Et0/1               Root FWD 100       128.2    Shr
Et0/2               Altn BLK 100       128.3    Shr
Et0/3               Desg FWD 100       128.4    Shr Edge
```

Una vez realizadas las pruebas, podemos confirmar que la red ofrece redundancia, garantizando alta disponibilidad y tolerancia a fallos; convergencia rápida ante la caída de un enlace gracias a RSTP; seguridad del tráfico mediante VLANs independientes; eficiencia y estabilidad con puertos bloqueados que evitan tormentas de broadcast; y escalabilidad, permitiendo la expansión de la red sin riesgo de loops.